ResearchAI: Deep Technical Architecture & Workflow Blueprint

Exhaustive Engineering Blueprint • Multi-Agent Orchestration • 8+ Academic Engines • 3-Layer Security

System Abstract: ResearchAI is a distributed multi-agent intelligence platform designed for deep literature synthesis, scientific discovery, and automated academic report generation. Rather than relying on a single generalist LLM prompt, ResearchAI employs a hierarchical multi-agent architecture with a centralized orchestrator (Mother Agent) and specialized child agents (Biomedical, Academic, Monograph/Patent, Web). The system concurrently queries 8+ global academic databases, normalizes hundreds of citations, extracts empirical findings, calculates consensus metrics, and streams publication-quality intelligence reports in multiple enterprise export formats.

1. Comprehensive Technology Stack & Infrastructure

Subsystem	Technology / Library	Granular Role & Implementation Details
Frontend Client	Next.js 14 (App Router) + React 18 + TailwindCSS + Lucide Icons	Server/Client components, responsive dark/light styling, interactive 6-tab ReportPreview, streaming message parser, and export trigger handlers.
State Management	Zustand (TypeScript Stores)	Reactive client stores for auth session (■uthStore), chat streaming (chatStore), bookmark management, and report state synchronization.
Backend REST API	FastAPI 0.115+ + Uvicorn + Pydantic v2	Asynchronous non-blocking ASGI server, strict Pydantic model validation, dependency injection, and automatic OpenAPI schema generation.
LLM Brain	Google Gemini 3.6 Flash (google-generativeai)	Primary neural reasoning engine with 8,192 token output capacity, sub-query decomposition, literature evaluation, and structured JSON synthesis.
Database & ORM	**SQLAlchemy 2.0 (Async/Sync) + SQLite / PostgreSQL**	Asynchronous database sessions (AsyncSession), UUID primary keys, soft-deletion mixins, relationship cascades, and connection pooling.
Caching & Rate Limit	Redis 7+ / In-Memory Fallback Cache	Sliding-window IP rate limiting, JWT refresh token revocation storage, query deduplication caching, and distributed worker pub/sub.
PDF Generation	ReportLab 5.0.1 (Pure Python Engine)	Custom two-pass NumberedCanvas (Page X of Y), fluid paragraphs avoiding table splitting issues, running headers/footers, and clean typography.
Document Exports	python-docx + markdown2	Automated generation of styled Microsoft Word .docx documents with table borders, formatted Markdown .md, and standalone HTML files.
Mail Dispatcher	Python smtplib + Gmail SMTP (App Password)	Asynchronous welcome emails and notification dispatch with styled HTML templates and background task execution.
Security Layer	■crypt + python-jose (HS256) + cryptography	3-Layer compound password hashing (HMAC-SHA512 + SHA-256 + 12-round Bcrypt), JWT access/refresh token pairs, and PKCE OAuth 2.0.

2. Hierarchical Multi-Agent Architecture & Orchestration Flow

ResearchAI eliminates LLM hallucinations and data gaps by separating the research task across five specialized agents: 1. **Mother Agent (Orchestrator):** Receives the user inquiry, classifies domain intent, formulates Boolean search queries, executes parallel dispatch across child agents, filters duplicates, and compiles the final structured report. 2. **Academic Literature Agent:** Queries OpenAlex, arXiv, Semantic Scholar, CORE, and DOAJ to harvest peer-reviewed papers, preprint benchmarks, and citation statistics. 3. **Biomedical & Clinical Agent:** Queries PubMed, Europe PMC, and WHO IRIS to extract clinical trial results, medical evidence levels, and therapeutic methodologies. 4. **Books, Patent & Policy Agent:** Queries Google Books, USPTO Patent data, and international databases for foundational literature, monographs, and intellectual property. 5. **Web & News Aggregator Agent:** Uses a multi-engine scraper fallback (DuckDuckGo, Wikipedia, Bing, Internet Archive) for recent breakthroughs and industry developments.

3. Integrated Scholarly & Academic Databases (8+ Engines)

Database	Coverage Scope	API Protocol & Rate Optimization	Extracted Metadata Fields
OpenAlex	250M+ cross-disciplinary publications, authors, institutions	REST API with polite email pool rate acceleration	Title, abstract, DOI, publication date, cited-by count, open access URL
PubMed / NCBI	35M+ biomedical, life science, and clinical trial studies	NCBI E-utilities API (esearch + esummary JSON/XML)	PMID, authors, journal title, study type, full abstract, MeSH terms
arXiv	2M+ preprints in Physics, Mathematics, CS, Quantitative Biology	arXiv API Atom/XML query parsing	arXiv ID, title, summary, authors, primary category, PDF link
Semantic Scholar	200M+ academic literature with AI-extracted summaries	Semantic Scholar Academic Graph REST API v1	PaperId, title, TLDR summary, citation count, influential citations
Europe PMC	40M+ life sciences articles, PMC full texts, preprints	Europe PMC RESTful Web Service	pmcid, journal, publication year, citedByCount, author details
CORE & DOAJ	Global open access research repositories and journals	CORE API v3 & DOAJ Search API	Repository link, full text download link, peer review status
Google Books	Monographs, foundational textbooks, academic volumes	Google Books Volumes REST API with Cloud API Key	Book title, authors, publisher, publishedDate, preview link
MultiSearch Web	DuckDuckGo, Wikipedia, Bing, Internet Archive, Mojeek	Aggregated MultiSearch fallback scraper integration	Snippet, source title, web URL, page summary

4. Enterprise Security & Cryptographic Architecture

1. 3-Layer Compound Password Hashing (Defense-in-Depth): Unlike standard systems using single-pass bcrypt, ResearchAI applies a 3-layer cryptographic transformation: • **Layer 1 (HMAC-SHA512 with Server Pepper):** The plain password is keyed with the private SECRET_KEY. This eliminates rainbow table precomputation and ensures that even if the SQL database is compromised, offline cracking is mathematically impossible without the server pepper. • **Layer 2 (SHA-256 State Digest):** Normalizes entropy and eliminates bcrypt's legacy 72-byte truncation limit, safely accommodating passphrases of arbitrary length. • **Layer 3 (Adaptive Salted Bcrypt 12 Rounds):** Applies 4,096 cost iterations with a cryptographically secure random salt, forcing significant CPU cost per attempt to resist GPU/ASIC brute-force attacks. • **Result Format:** Stored in the database as 2b$... with automatic backward compatibility for legacy accounts. **2. Google OAuth 2.0 & Identity Security:** • Implements Google Identity Services with client ID and secret exchange. • Direct ID token credential verification via Google's secure tokeninfo endpoint. • Automatic user creation, profile picture linking, and session token generation upon Google authorization. **3. JWT Session Tokens & Redis Revocation:** • Dual-token architecture: 60-minute short-lived ■ccess_token for API authorization + 7-day efresh_token stored in Redis with revocation checks on /auth/logout. • IP-based brute-force rate-limiting on registration (5 attempts/hour) and login endpoints.

5. Report Synthesis & Multi-Format Document Generation Engine

Synthesis Pipeline: Raw retrieved documents are normalized, citation-keyed, and submitted to Gemini 3.6 Flash with strict JSON Schema constraints. The model outputs 6 structured sections: 1. **Executive Summary & Foundational Context:** High-density briefing with methodology verification protocols. 2. **Empirical Findings:** Numbered technical findings, quantitative metrics, key takeaway pills, and supporting evidence tags. 3. **In-Depth Thematic Analysis:** Granular examination of underlying mechanisms, trade-offs, and systemic impacts. 4. **Evidence Matrix & Comparative Stances:** Multi-dimensional comparison of contrasting scientific perspectives and benchmarks. 5. **Conclusions, Limitations & Strategic Roadmap:** Consensus milestones, methodological constraints, and high-impact future directions. 6. **Scholarly Bibliography:** Complete numbered APA-style reference list with title, author, year, venue, and clickable DOI/URLs. **Document Generation Engine (eport_service.py):** • **PDF Generation:** Built with ReportLab 5.0.1. Employs a custom two-pass NumberedCanvas that counts exact total pages, running headers on pages 2+, left-accented emerald callout boxes, and flowing paragraphs that eliminate blank page gaps. • **Word (.docx) Generation:** Built with python-docx, generating native XML headings, custom callout tables, and styled bulleted lists. • **Markdown (.md) & HTML (.html):** Clean semantic markdown AST and standalone responsive HTML with embedded CSS styling.

6. Email Dispatch & Automated Notification Pipeline

Email Dispatch Service (email_service.py): • Integrated with Gmail SMTP (smtp.gmail.com:587) using TLS encryption and App Passwords. • Asynchronous non-blocking dispatch via asyncio.to_thread ensuring API response latency is zero. • Dispatches branded HTML welcome notifications and account confirmations immediately upon registration and Google OAuth sign-up.